

7 Polar coordinates	7.1	Understand and use polar coordinates and be able to convert between polar and Cartesian coordinates.	
	7.2	Sketch curves with r given as a function of θ , including use of trigonometric functions.	The sketching of curves such as $r = p \sec(\alpha - \theta)$, $r = a$, $r = 2a \cos\theta$, $r = k\theta$, $r = a(1 \pm \cos\theta)$, $r = a(3 + 2 \cos\theta)$, $r = a \cos 2\theta$ and $r^2 = a^2 \cos 2\theta$ may be set.
	7.3	Find the area enclosed by a polar curve.	Use of the formula $\frac{1}{2} \int_{\alpha}^{\beta} r^2 d\theta$ for area. The ability to find tangents parallel to, or at right angles to, the initial line is expected.